

Dr. Siva K. Balasubramanian

### Description of Research Project Paper – MSC 612 – Spring 2025

***This Research Project seeks to provide you a useful learning experience to execute the empirical estimation, perform additional analyses, interpret and write up research results in insightful manner following the use of the SEM Research Method to a real dataset.***

***This Research Project is due on April 30, 2025 by 10 pm.*** Please email project files (including assignment, program files, datasets and outputs) to [sivakbalas@stuart.iit.edu](mailto:sivakbalas@stuart.iit.edu), [rxu21@hawk.iit.edu](mailto:rxu21@hawk.iit.edu), and also submit your files to BlackBoard. **Student Oral Presentations on their respective Research Projects scheduled on May 7 2025 as per the syllabus.**

Groups for Research Project:

Group A

Group B

Group C

Group D

Group E

Important Note: Please remember that all your datafiles, survey questionnaire file, and baseline model file (and other pertinent files) for the Project your group has been assigned is available on Onedrive subfolder titled “Research Project – MSC 612....” for your assigned Group as shown above.

We will meet over zoom beginning March 6 at 2 pm as per the schedule below:

The zoom link that we will use for all our Research Project related online meetings is as follows:

<https://lewisu-edu.zoom.us/j/6174917157?pwd=YzY2Z2cxVWF5TlY1WTlrZ094c3Z2QT09&omn=89884768955&from=addon>

- A. March 6 (Thursday) at 2 pm – General Session that will provide you access to all data, baseline model and other resources for the Research Project
- B. March 11 and 13 at times shown below for each group:

Group A: between 2 pm and 2:20 pm.

Group B: between 2:20 and 2:40 pm

Group C: between 2:40 and 3:00 pm

Group D: between 3:00 and 3:20 pm

Group E: between 3:20 and 3:40 pm

- C. We will follow the broad schedule above for our individual group meetings during all other weeks with the exception of Spring Break week.

Please note that, after the March 13 meeting, we will next meet only on March 27 and we will follow the above schedule for all meetings after that.

Generally, a guiding principle I use in choosing datasets for your Research Project paper is that they should contain at least 25 variables (manifest or observed variables also called indicators) with at least 5 response options each (typically these options could be in the Likert format i.e., strongly disagree (1), disagree (2), neither disagree or agree (3), agree (4), strongly agree (5) that can be coded with respective values as shown – we always use the coding convention of values increasing from left to right such that increasing values reflect more positivity; or they could be semantic differential such as ugly \_\_\_ \_\_\_ \_\_\_ \_\_\_ beautiful scale with coded values reflecting 1 through 5 in that range).

I also prefer to work with a dataset with at least 1000 observations, so that after addressing any issues due to a decrease in sample size caused by missing values (SEM does not accommodate missing values), we are left with a reasonable sample size to do calibration/validation analyses and multiple group analyses. Typically, the models I develop involve about 3-7 latent constructs and about 20-40 indicators.

Later today, I will share an individualized, project-specific Google Drive folder with each of you. I have requested Ruiqing/Zheng to populate the following files in that folder and expect that task will be completed by the last week of March 2025:

1. The SPSS datafile and questionnaire for the project assigned to you from PEW website originally, as well as cleaned files (the cleaned files take care of rearranging/recoding response options such that for a given construct, higher numbers on the measurement scale convey higher levels of the construct being measured.)
2. A CFA/SEM model that I have developed for the project assigned to you (This is referred to as Project X Baseline Model.pdf where X represents one of the following alphabets assigned to your project: A, B, C, D, E, or F.) More details of student specific assignments of projects/datasets extracted (and cleaned) from the PEW website are given in the accompanying Word document.
3. The PEW website usually has several press releases and other published reports following their analyses of the dataset for your project. I request you to read those releases/reports carefully for potential citations in your research report as appropriate.
4. My expectations for your Research Project report follow:
  - Use the Book Chapter file (Brodowsky doc, Chapter 7, pages 83-103 that my co-authors and I wrote) as a template document to prepare your multiple group analyses (MGA) portion of your Research report. You should pay particular attention how each path in your Project's Baseline Model represents a theoretical hypothesis, so I expect you to conduct library research and develop appropriate

theoretical arguments that support the direction and sign of that hypothesis, along the lines of what the Chapter 7 above does. In addition, your report should use the Novak et al 2001 article (discussed and replicated in class) as another template to prepare your Research Report. Your report *should look like a manuscript that can potentially be submitted to an academic journal*. Since we have discussed several academic journal articles/manuscripts in class, you should be quite familiar with that “look.” Specifically, it should have all of the following distinct sections: Introduction, Literature Review/Hypotheses (this should present/explain each model path in the baseline model and explain the theory/rationale for that path), Data description/source, CFA/SEM analyses description, Multiple Group Analyses, Validation analyses, Discussion, Contributions/Implications and Limitations, and Future Research Directions. More specific issues/guidance to develop and present each section of your Research Project Report follows:

- **Introduction.** Provide a descriptive set of paragraphs motivating why your research is interesting, along with topical/background references to justify or establish rationales that support your statements.
- **Literature Review/Hypotheses:** The CFA/SEM model that I provided for your project originally, along with any modifications to that in the CFA/SEM estimation process becomes the basis of this section. Each path in the SEM model that you describe in this paper represents a hypothesis, so you need to research (I recommend EBSCO at IIT’s Galvin library website) academic literature related to that inter-relationship among constructs corresponding to that path and motivate your hypothesis accordingly. I strongly recommend consulting the Novak, Hoffman et al (2000) paper as well as my Book Chapter document in developing the literature review/hypotheses section.
- **Data description:** the PEW project website documents related to your dataset will generally provide an overview that you will find useful here. Describe the dataset, attribute the source of data (who collected it and when etc.) and then describe how you cleaned it. (although the datasets that are provided to you are already cleaned, this should be explained in every research article).
- **CFA/SEM analyses.** You need to estimate the model and the PEW data+other documents that I provided you using the same template that you used for replicating the Hoffman/Novak paper in class. Prepare tables in the same format as used in the Hoffman/Novak paper to summarize your CFA/SEM analyses.
- **Multiple Group analyses.** I may identify some Grouping variables for your specific project when I email you the questionnaire/other docs for your Research Project (these MGA variables are identified at the bottom of your Project’s baseline model page. If that is the case, use these Group variables to run multiple group analyses in MPlus.
- **Validation analyses.** You may recall that Hoffman/Novak divided their sample into calibration and validation samples. Finish all your analyses first on the entire sample. Once those results look reasonable, please divide your sample into calibration and validation samples and revisit the CFA/SEM analyses in the manner that Hoffman/Novak illustrate.

- **Discussion.** This section should relate your results to state how your hypotheses testing proceeded. For this section and the next two sections below, please use my book chapter document (Word doc attached to the email that accompanied this doc) as a template for preparing your report.
- **Contributions/Limitations**
- **Future Research Directions**

5. Individual guidance sessions with Professor. I plan to devote at least 4 hours/week (between 2-4 pm on Tuesdays and Thursdays in April 2025. Both Zheng/Ruiqing will also attend those sessions based on their availability. The goal of these individual sessions is to provide any instruction/guidance you need with MPlus analyses for CFA/SEM, MGA and validation analyses. The schedule for these guidance sessions are in 7 below.
6. As always, I am only an email away for help on any aspect of your research project. As I mentioned in class, my goal is to work with each of you through one-to-one meetings during this semester and after the semester to develop your research project into a submission for a journal. Hopefully, we will have a good outcome on that goal. Good luck with your research project!
7. Research Project Assignments to each student will be announced later. Each student will be assigned one of the following projects. Regular zoom instruction session with each student will be announced on a weekday at the following times:

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|--------------------------------------------------------------------------------------|--------------------|
| <b>Project A – Twitter Survey 2018 (Trump) –</b>                                     | 2 pm and 2:20 pm   |
| <b>Project B – Teen &amp; Privacy Management 2012 -</b>                              | 2:20 pm to 2:40 pm |
| <b>Project C – Teen Relationship Survey 2014/2015 -</b>                              | 2:40 pm to 3 pm    |
| <b>Project D – Teens &amp; Tech Survey 2018 -</b>                                    | 3:00 pm to 3:20 pm |
| <b>Project E – American Trends 2020 – Wave 63</b>                                    | 3:20 pm to 3:40 pm |
| <b>Project I – Science Survey – Oct 1, 2019-March 15, 2020 (Dataset= I – Russia)</b> |                    |